

Netflix Content Analytics

Content Trends • Audience Signals • Catalog Insights

Professional Project Report

Project Overview

This project analyzes a Netflix catalog dataset to understand content composition, release trends, genre and country distribution, ratings, popularity signals, movie runtime, and show seasons. The workflow combines Python, SQL, Excel, and Power BI to turn catalog metadata into an interactive analytical story.

1. Business Objective

The analysis focuses on catalog-level questions around content mix, release trends, genres, production countries, ratings, popularity, movie duration, and show season patterns.

2. Dataset & Scope

Source: enriched Netflix catalog dataset from the GitHub repository **amirtds/kaggle-netflix-tv-shows-and-movies**. The source catalog contains title metadata including type, release year, certification, genres, production countries, runtime, seasons, IMDb measures, and TMDb measures. It represents a U.S. Netflix catalog snapshot as of May 2022; therefore, the results should not be described as a current 2026 Netflix catalog or as industry-wide OTT statistics.

3. Data Preparation

- Removed the single record with a missing title.
- Treated zero runtime values as missing for duration analysis.
- Filled missing age certification values with “Not Rated”.
- Parsed genre and production-country list fields.
- Created release decade and content age features.
- Created a high-rated flag using IMDb score ≥ 7 .
- Final cleaned dataset: 5,805 titles and 22 analytical columns.

4. Key Metrics

Metric	Value
Total Titles	5,805
Movies	≈3,759
Shows	≈2,046
Average IMDb Rating	≈6.53
High-Rated Titles (IMDb ≥ 7)	≈2,064

5. Power BI Dashboard

- Content Mix
- Content Growth Over Time

- Top 10 Genres
- Top 10 Production Countries
- Average IMDb Rating by Content Type
- IMDb Rating Distribution
- IMDb Rating vs TMDB Popularity
- Content Distribution by Release Decade
- Movie Runtime Distribution
- Show Seasons Distribution

6. Interactivity

The dashboard includes slicers for Content Type, Release Year, Age Certification, and Release Decade, plus a Reset Filters button for returning to the default dashboard state.

7. Business Insights Framework

The dashboard supports evidence-based analysis of catalog composition, content growth, genre and geographic concentration, rating quality, and rating-versus-popularity relationships. Exact findings should be read from the dashboard under the relevant filter context rather than inferred beyond the dataset.

8. Tools & Technologies

Python • Pandas • Matplotlib • Seaborn • SQL/MySQL • Excel • Power BI • GitHub

9. Limitations

IMDb and TMDB measures are third-party signals and are not direct Netflix engagement metrics. Genre and country fields can contain multiple values per title. The dataset does not provide viewing hours, completion rate, subscriber behavior, revenue, or current catalog availability.

10. Conclusion

This project demonstrates an end-to-end analytics workflow: data cleaning and feature engineering in Python, analytical querying in SQL, spreadsheet validation in Excel, and interactive storytelling in Power BI. It provides a reusable framework for exploring Netflix catalog trends and audience-related signals while keeping conclusions tied to the available data.